

Green Metals

Demand revolution gathers momentum



Green demand dominates fundamentals. Whilst many investors can see the structural demand surge for green metals (aluminium, copper, nickel, lithium and cobalt), the true impact it has on fundamentals remains underappreciated in the market today. Already, the acceleration of investment in electric vehicles, renewable power generation and energy storage has helped metals demand overcome a property downturn in China and cyclical slowdown in the West. And yet, this green demand surge has only just begun. In copper, green demand accounted for 4% of consumption in 2020, 7% today and is set to rise to 17% by 2030, accounting for 47% of additional copper demand (2023-30E), equivalent to China's demand share in the 2010's (45%, 2010-2020). Should policy continue to converge to a "Net-Zero Emissions" path, the world would need an additional 54% copper by 2030 on top of our base case, and an estimated 198Mt of cumulative green demand by 2050, equivalent to the last 10 years' aggregate copper mine supply. Green demand is at the heart of the metal supercycle and should be top of mind for investors – without it, copper would face a multi-year surplus with peaking Chinese property demand.

China still drives demand in 2023. China will continue to be the centre of green demand in 2023 as solar installations and pent-up EV sales leave it accounting for more than half of global green metals demand. Yet policy is helping diversify these green demand drivers as RePowerEU and the Inflation Reduction Act (IRA) cement clean-tech as the centre piece for Western reindustrialisation over the coming decade, softening the impact a slowdown in cyclical demand channels. In 2023, we expect this to generate 2.6Mt of copper consumption (8% of global demand) and 5.1Mt of aluminium (7% of global demand) as green sectors dominate, accounting for 58% of demand growth for copper and 65% for aluminium. After a rapid expansion in 2022, battery metal demand will face a drag from slowing onshore battery capacity expansions, as green demand for lithium grows at 50%, nickel at 35% and cobalt at 32% in 2023, in our view. Finally, we expect silver green demand to grow by 30% in 2023 as global solar capacity ramps up.

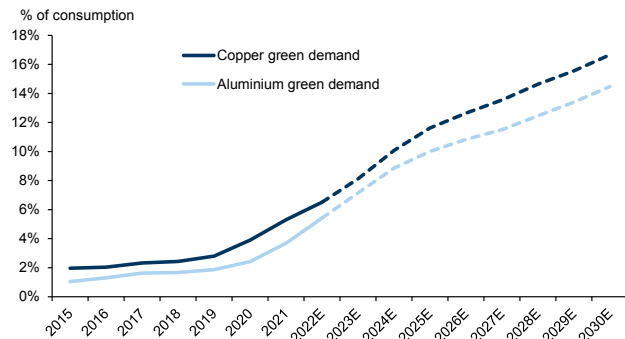
Geopolitical competition is going green. The most important development over the last year has been the rise in strategic industrial competition between China, Europe and the US as each race attract clean-tech capital. The US IRA's protectionist elements catalysed European policymakers to launch the "Net Zero Industry Act" (NZIA) outlined in the Green Deal Industrial Plan, in addition to their prior flagship

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policy, RePowerEU. Crucially, we are now seeing policy support in one region incentivising further subsidies and support in others. Indeed, the US – previously lagging in the adoption of clean-tech – is set to see an acceleration of renewable installation and EV penetration. The Inflation Reduction Act alone from 2023 to 2030 will add an estimated 180kty (~1% of global demand) of copper demand, 530kty (~1% of global demand) of aluminium, and 60kty LCE (~1% of global demand) of lithium demand. Should Europe meet the RePowerEU proposals, targeted 2030 renewables capacity would rise by 15% boosting average annual EU demand growth rates for copper and aluminium by 1.5% and 2% respectively in the next 7 years, exemplifying the additional demand upside from strategic clean-tech re-industrialisation.

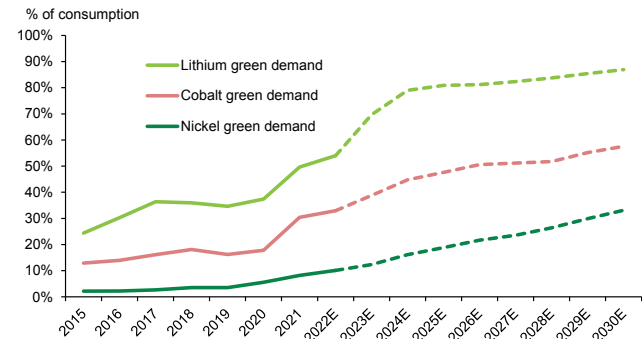
Green Metals Demand in 16 charts

Exhibit 1: Green demand will represent 47% of additional copper demand for rest of the decade



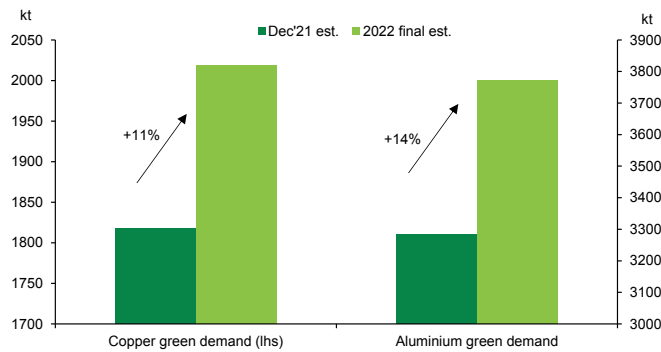
Source: ICA, IEA, BNEF, Goldman Sachs Global Investment Research

Exhibit 2: Lithium green demand will constitute >90% of total demand by 2030



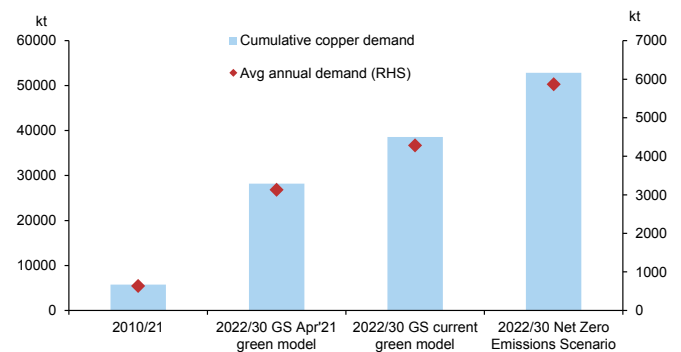
Source: IEA, BNEF, Goldman Sachs Global Investment Research

Exhibit 3: Higher renewables installation and strong Chinese EV sales have brought green demand forward



Source: Goldman Sachs Global Investment Research

Exhibit 4: Policy convergence to a "Net-Zero Emissions" path will require an additional 54% copper, far surpassing our base case Green demand from solar, wind & EVs

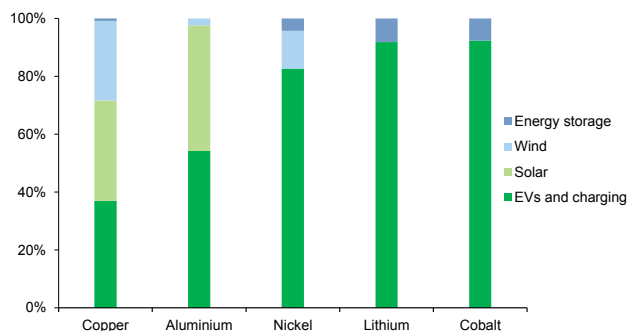


The IEA Net Zero Emissions by 2050 Scenario is a pathway for the global energy sector to achieve net zero CO2 emissions by 2050.

Source: IEA, Goldman Sachs Global Investment Research

Exhibit 5: Electric vehicles have been the driver of green metal demand

2022 green demand, sector share



Source: Goldman Sachs Global Investment Research

Exhibit 6: Green demand growth resilient into 2023

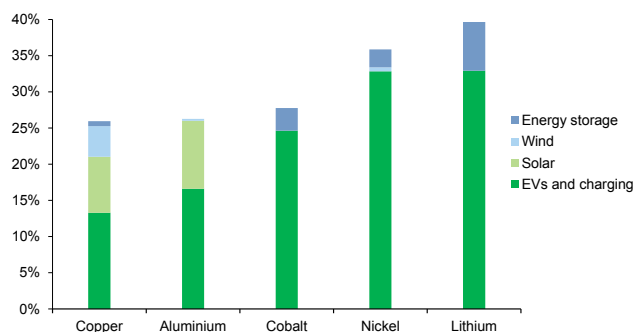
	Copper		Aluminium		Lithium*		Nickel		Cobalt	
	22E	23E	22E	23E	22E	23E	22E	23E	22E	23E
EU	447	484	692	827	46	64	36	44	7	8
US	208	257	380	533	25	51	20	31	4	6
China	949	1486	1968	3115	271	401	190	290	39	51
Global green demand	2019	2599	3773	5124	406	609	296	400	55	72
as % of global demand	7%	8%	5%	7%	66%	74%	13%	16%	33%	39%
YoY growth	25%	29%	49%	36%	52%	50%	25%	35%	14%	32%
Incremental volume	404	580	1234	1351	138	204	59	104	7	18
as % of global demand	1%	2%	2%	2%	18%	23%	2%	3%	4%	9%

* lithium is in kt LCE

Source: Goldman Sachs Global Investment Research

Exhibit 7: Copper's green demand growth has the most diverse set of drivers

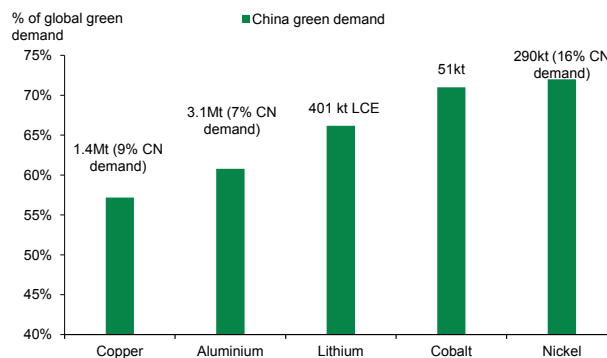
Based on GSe green demand growth over 2023-25E



Source: Goldman Sachs Global Investment Research

Exhibit 8: China represents 50-70% of green metals demand across the complex

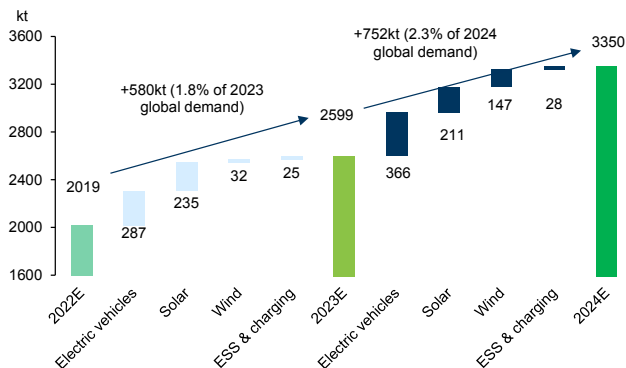
2023 China share of global green demand & China green demand volumes, in kt*



*Lithium demand volumes are in kt LCE

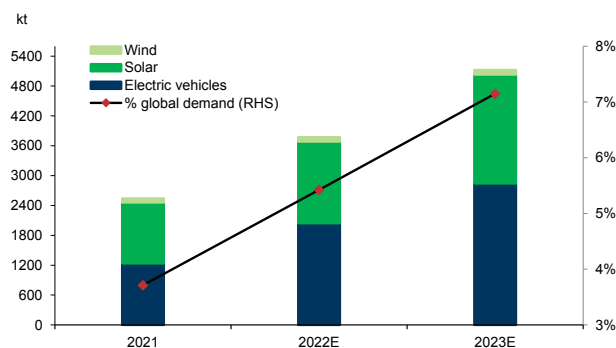
Source: Goldman Sachs Global Investment Research

Exhibit 9: Electric vehicles and solar installations are the main drivers of green copper demand over the next 2 years
Green copper demand GSe ('000 tonnes)



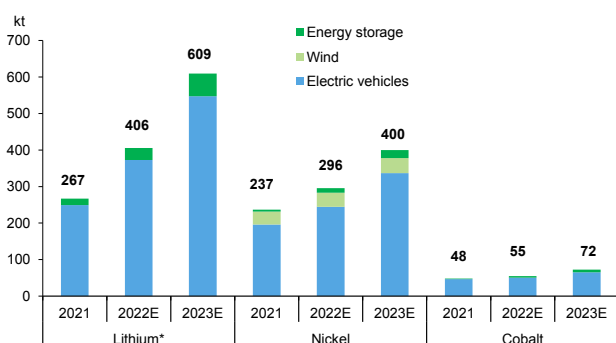
Source: Goldman Sachs Global Investment Research, BNEF, Company data

Exhibit 10: We estimate 2023 green aluminium refined demand to grow to 5Mt (7% of global demand)
GS Aluminium Green Demand Model



Source: Goldman Sachs Global Investment Research

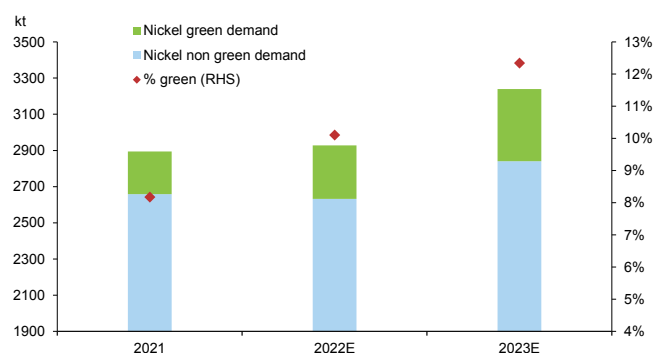
Exhibit 11: Rapid growth in battery metal demand as EV uptake accelerates
Battery metals green demand



*Lithium is in kt LCE

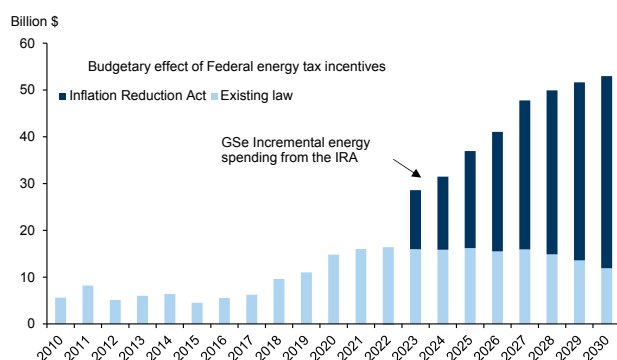
Source: Goldman Sachs Global Investment Research

Exhibit 12: For nickel, despite accounting for 12% of total demand, green sectors accounts for 33% of growth into 2023



Source: Woodmac, Goldman Sachs Global Investment Research

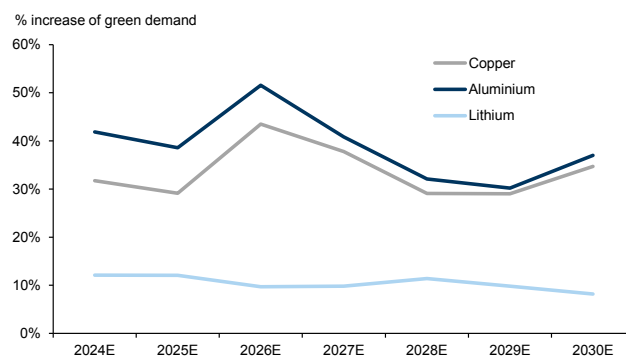
Exhibit 13: The Inflation Reduction Act is estimated to increase energy spending by \$264 bn over the next 8 years



Source: Goldman Sachs Global Investment Research

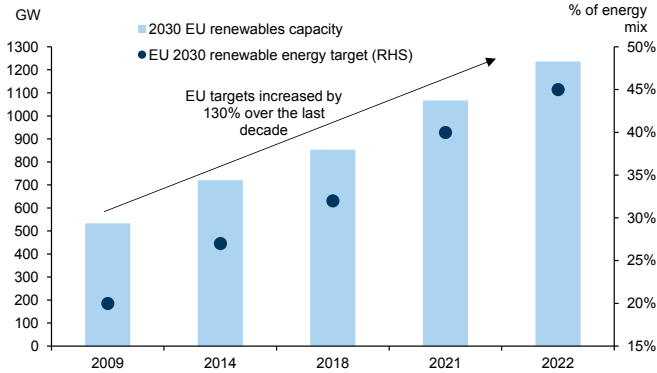
Exhibit 14: The US is set to see an acceleration of renewable installation and EV penetration, inducing material upgrade to green demand

IRA metals demand boost as % of green demand



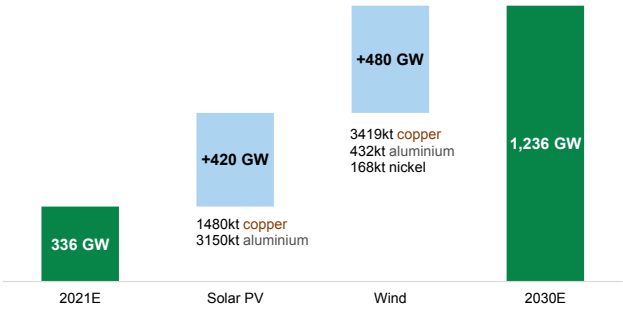
Source: Goldman Sachs Global Investment Research

Exhibit 15: Increasing EU 2030 renewables target has boosted green metals demand
2030 EU renewable energy target (%) and implied renewables capacity (RHS)



Source: European Commission, Goldman Sachs Global Investment Research

Exhibit 16: EU's renewable installments outlined by RePowerEU this decade will require c.5Mt tonne of copper and 3.5Mt of aluminium



We assume the required 1236GW needed for 45% renewable share are achieved with solar & wind

Source: Goldman Sachs Global Investment Research, IRENA, IEA, BNEF

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Quantifying Green Metals Demand

The surge in green transition demand has continued to offset the drag from non-green channels. In 2022, the growth in green demand for copper entirely offset the drag from the slowdown in property in China. For copper, green demand rose 25% to 2Mt (7% of global copper demand vs. 4% in 2020) whilst aluminium green demand rose to c.4Mt (5.4% of global demand), accounting for all primary aluminium demand growth globally. The drivers of green demand growth are strong electric vehicles sales and accelerated renewables installations, in particular solar (see [Exhibit 18](#)). Amongst battery metals, green lithium demand increased by 52% to 406kt LCE in 2022 (53% of global demand) due to the rapid rise of LFP battery deployment in China. We calculate global green nickel and cobalt demand in 2022 to be at 296kt (10% of global demand) and 55kt (33% of global demand). The upside surprise in green demand last year was a result of higher-than-expected Chinese EV sales, Beijing's strong focus on the renewables space and improving economies of scale for the solar and wind supply chains. China's push for electrification has been shielded from overall weak economic growth, with EV sales almost doubling y/y in 2022 (+91% y/y), solar installations up 51% y/y and wind up 9% y/y. For 2023, we expect green demand for copper to grow at 30% to 2.6Mt (8% of global demand) and aluminium at 36% to 5.2Mt (7% of global consumption). Amongst battery metals, we see green demand for lithium to grow at 50%, nickel at 35% and cobalt at 32% (see [Exhibit 17](#)). Globally, we estimate this structural rise in green demand to near outweigh the cyclical slowdown and expect ~3% growth in copper and aluminium consumption globally. This impact of green demand will continue to strengthen over the decade through increasing share of green sectors in global consumption (see [Exhibit 25](#)) backed by the push for energy security and the green transition.

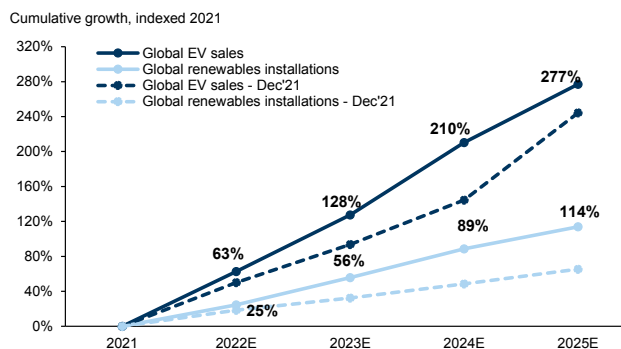
Exhibit 17: Green demand growth resilient into 2023

	Copper		Aluminium		Lithium*		Nickel		Cobalt	
	22E	23E	22E	23E	22E	23E	22E	23E	22E	23E
EU	447	484	692	827	46	64	36	44	7	8
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* Lithium is in kt LCE

Source: Goldman Sachs Global Investment Research

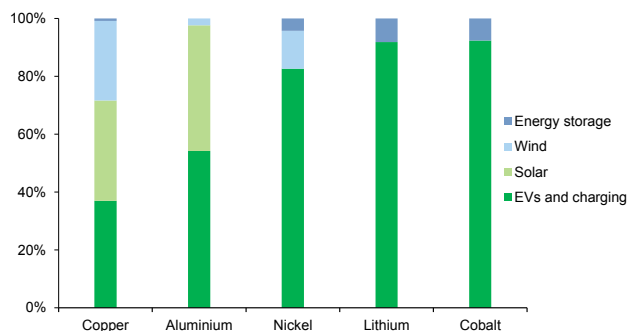
Exhibit 18: Strong Chinese EV sales, strong policy focus on the renewables and improving economies of scale will support green demand



Source: Goldman Sachs Global Investment Research, BNEF, IEA, IRENA, Wind

Exhibit 19: Aluminium, copper and nickel set to benefit from both EVs and renewables

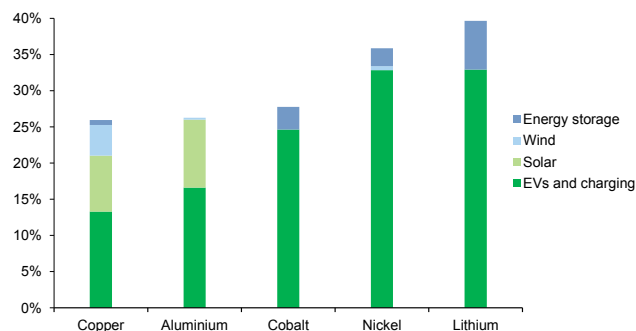
2022 green demand, sector share



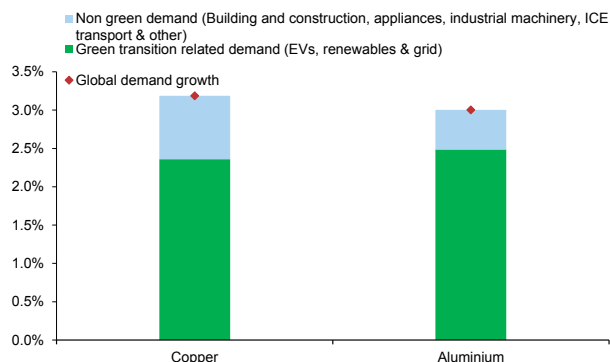
Source: Goldman Sachs Global Investment Research

Exhibit 20: Copper's green demand growth has the most diverse set of drivers

Based on GSe green demand growth over 2023-25E



Source: Goldman Sachs Global Investment Research

Exhibit 21: Green transition related demand is the offset to a cyclical slowdown into 2023


Source: Goldman Sachs Global Investment Research

Exhibit 22: The green demand share has continued to increase across base and battery metals

	Share of green demand in global consum				
	2020	2021	2022E	2023E	2025E
Aluminium	3%	4%	5%	7%	10%
Copper	4%	5%	7%	8%	12%
Nickel	36%	50%	53%	69%	80%
Lithium	18%	30%	33%	39%	48%
Cobalt	6%	8%	10%	12%	19%

Source: Goldman Sachs Global Investment Research, BNEF, ICA, Wind, Bloomberg

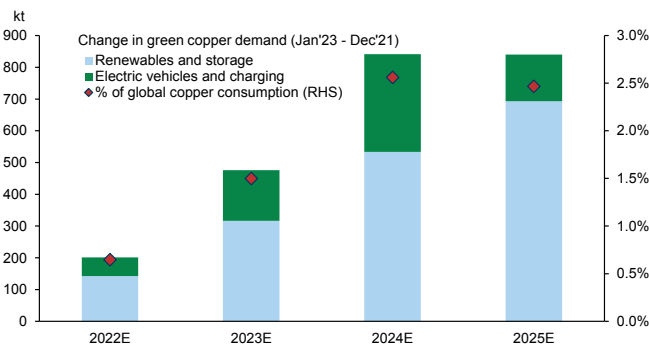
Copper: Solar installations lifting green demand higher

The shift away from economies built on hydrocarbons to a more sustainable one cannot happen without copper - which is essential for electromobility, wind and solar energy generation and energy storage. We estimate global green copper demand amounted to 2.02Mt (+204kt vs. our Dec'21 model) in 2022, rising +404kt driven by higher-than-expected Chinese EV sales and continued strength in solar deployments. For 2023, we expect copper green demand to grow c.30% to 2.6Mt, accounting for 58% of global demand growth, thanks to rising EV sales globally (+3.8mil, 40% y/y) and the acceleration of renewables, especially in China where our Equity analysts upgraded solar installations by 50GW and EV sales by 0.5Mil units (see Exhibit 24). Furthermore, green demand for copper has second-round effects on power-grid related demand. To support growing renewable energy installations, countries need digitalised power grids – ones which can monitor electrical loads' supply and demand in real time and are more copper intensive (see Exhibit 25), requiring 304kt (1% of global demand) metal in 2022 and 148kt in 2023. As a result of the rising demand from green channels, copper remains supported against a drag from recessionary pressures through cyclical sector consumption. We expect this trend to continue in 2023 and see global copper demand

growing c.3.2% y/y even as non-green demand ex-grid remains flat. Finally, the green transition will continue to be the main driver of copper consumption growth post 2023, in our view, with green demand expected to grow on average by 598kty in the period between 2024 and 2030, reaching 6.8Mt in 2030, accounting for 17% of global demand versus just 5% in 2021 (see [Exhibit 26](#)).

Exhibit 23: Policy and improving economies of scale lifted green demand in 2022 and estimates for years ahead

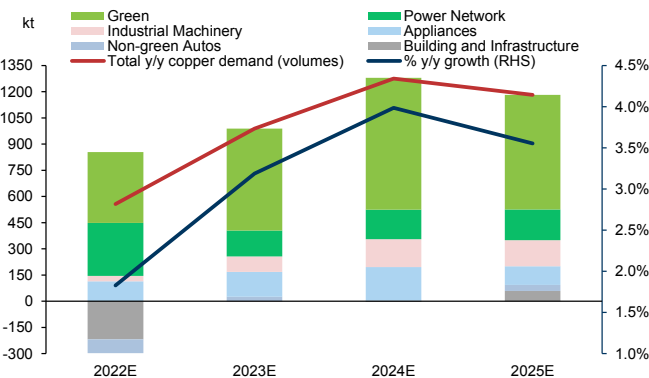
GSe green copper demand (Jan'23- Dec'21 model)



Source: Goldman Sachs Global Investment Research

Exhibit 25: Electrification will account for 60% of global demand growth over 2022-25E

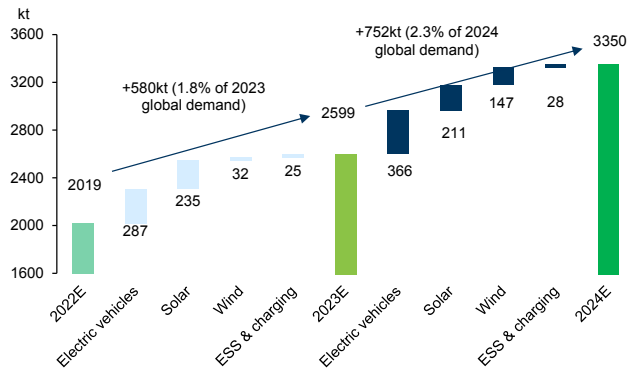
YoY global copper demand growth



Source: Goldman Sachs Global Investment Research

Exhibit 24: Electric vehicles and solar installations are the main drivers of green copper demand over the next 2 years

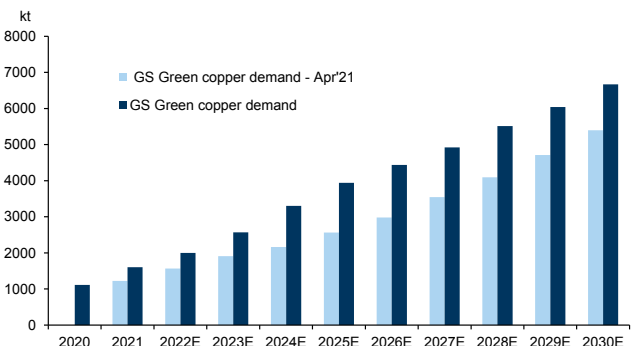
Green copper demand GSe ('000 tonnes)



Source: Goldman Sachs Global Investment Research, BNEF, Company data

Exhibit 26: Increasing policy commitment and improving economies of scale have increased our green estimates

Green demand from solar, wind & EVs

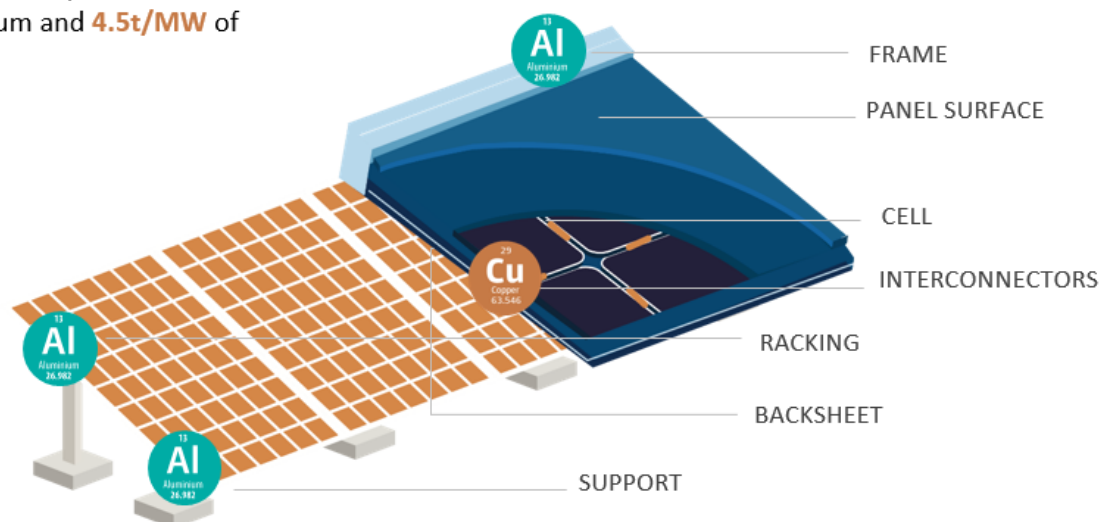


Source: Goldman Sachs Global Investment Research

Exhibit 27:

SOLAR PANELS

In a solar panel there are approximately **7.5t/MW** of aluminium and **4.5t/MW** of copper.



Source: World Bank, ICA, Copper Alliance, Goldman Sachs Global Investment Research

Exhibit 28: GS Copper Green Demand Model

GS Copper Green Demand Model										
'000 tonnes	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Solar	526	701	936	1147	1282	1383	1470	1561	1646	1733
Wind	619	554	587	733	919	1006	1145	1288	1349	1443
Stationary energy storage	10	17	31	46	59	70	81	89	101	115
Electric vehicles	441	718	1005	1371	1674	1961	2201	2538	2895	3313
Charging infrastructure	19	29	40	53	68	86	106	128	153	180
Global green demand	1615	2019	2599	3350	4002	4506	5002	5604	6143	6784
YoY growth	75%	25%	29%	29%	19%	13%	11%	12%	10%	10%
% global copper demand	5%	7%	8%	10%	12%	13%	14%	15%	16%	17%
Europe	357	447	484	607	759	1010	1227	1469	1653	1768
US	162	208	257	500	661	743	815	923	976	1150
China	773	949	1486	1775	2028	2136	2257	2402	2555	2720
RoW	324	414	373	469	553	617	703	811	959	1146

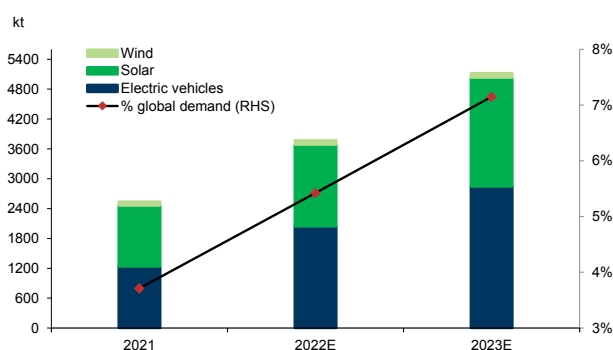
Source: Goldman Sachs Global Investment Research

Aluminium: Green demand will drive two-thirds of global demand growth in 2023

Similar to copper, the sharp increase of both EV and solar panel sales in China has boosted green aluminium demand as it is an important component in EVs and solar power supply. We expect green aluminium demand to have grown 49% (1.23Mt) in 2022, to 3.7Mt, accounting for c.6% of total demand (vs. 3% in 2020). Last year, we estimate China accounted for 67% of green aluminium demand growth (833kt), followed by Europe (+114kt) and the US (+116kt). Into 2023, we expect aluminium demand to expand at a slower rate than 2022, growing 36% y/y and reaching 5.1Mt

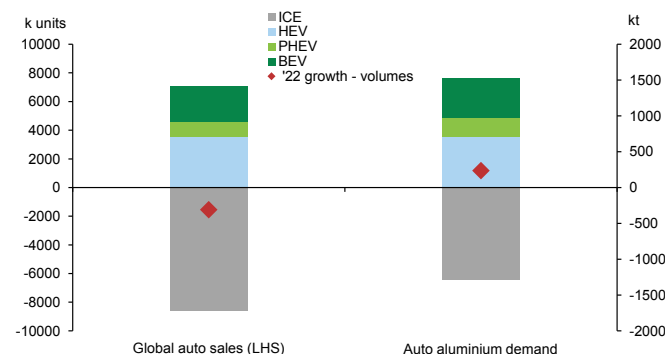
(see Exhibit 29), representing more than half of global demand growth (3%). The slowdown in demand growth is a result of moderation in EV sales growth. However, increasing EV penetration will continue to more than offset the demand decline from falling ICE vehicles sales due to their higher aluminium intensity (1.5 times ICE), as seen in 2022. In 2022, the growth in EV sales (+3.8mn units) equals half of the ICE sales decrease (-8.6mn units), yet due to a higher metal intensity of EVs, the net effect was positive for aluminium transport demand (+237kt, 0.5% of global demand) (see Exhibit 30). Overall, we estimate aluminium demand growth of 1.3% in 2022 and 3% in 2023 as a result of support from green demand. Over the long term, we expect green aluminium demand to rise to 12Mt by 2030 (14% of global consumption), growing on average by 1Mty (>1% of global demand) from 2024-2030.

Exhibit 29: We estimate 2023 green aluminium refined demand to grow to 5Mt (7% of global demand)
GS Aluminium Green Demand Model



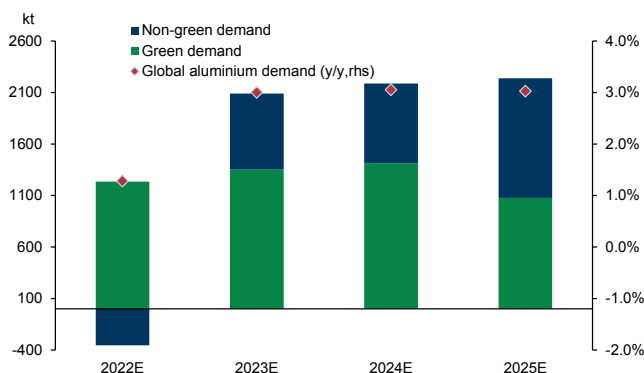
Source: Goldman Sachs Global Investment Research

Exhibit 30: Increasing aluminium demand from rising EV penetration, which is 1.5x aluminium intensive vs. ICE vehicles
2022 global auto sales & related aluminium demand growth, in volumes



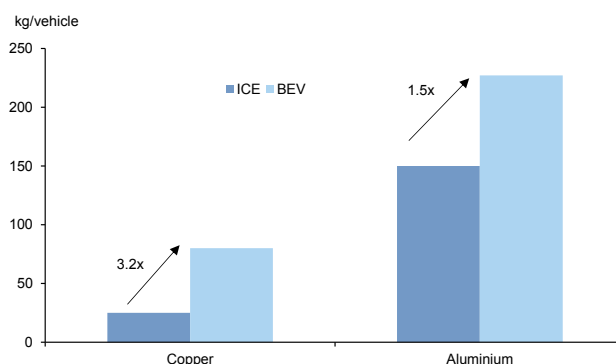
Source: Goldman Sachs Global Investment Research

Exhibit 31: Despite a cyclical slowdown in industrial sectors, aluminium demand remains supported by green demand



Source: Goldman Sachs Global Investment Research

Exhibit 32: Relatively lower green leverage for aluminium, as EVs are only 1.5x more aluminium intensive than ICE



Source: ICA, CRU, Goldman Sachs Global Investment Research

Exhibit 33: GS Aluminium Green Demand Model

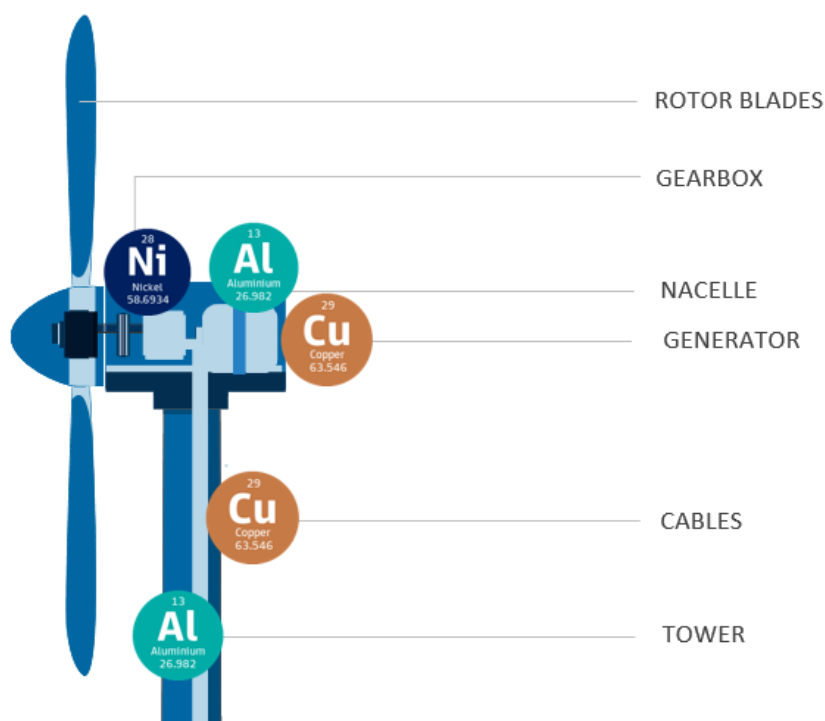
GS Aluminium Green Demand Model										
'000 tonnes	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Solar	1225	1640	2189	2693	3012	3247	3451	3670	3879	4090
Wind	82	88	94	110	131	144	159	174	183	195
Electric vehicles	1235	2045	2842	3736	4474	5074	5591	6333	7115	8036
Global green demand	2543	3773	5124	6539	7617	8465	9202	10176	11177	12321
YoY growth		48%	36%	28%	16%	11%	9%	11%	10%	10%
% of global refined demand	4%	5.4%	7.1%	9%	10%	11%	12%	12%	13%	14%
Europe	578	692	827	1077	1327	1746	1992	2359	2687	2878
US	265	380	533	987	1308	1501	1712	1872	2028	2325
China	1135	1968	3115	3586	3958	4056	4180	4399	4620	4922
RoW	565	733	649	889	1025	1162	1319	1546	1842	2196

Source: Goldman Sachs Global Investment Research

Exhibit 34:

WIND TURBINES

A 3 megawatts wind turbine contains up to **4.7t** of copper, **1t** of aluminium and **0.4t** of nickel



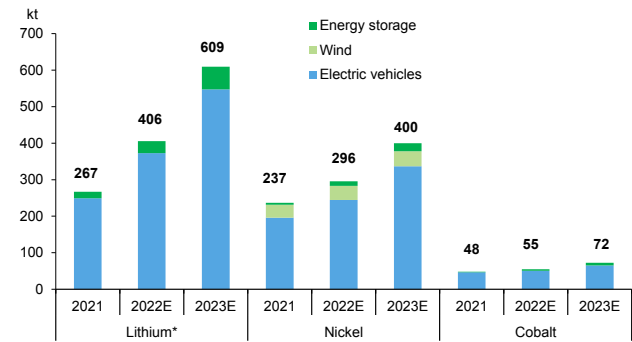
Source: World Bank, ICA, Copper Alliance, Goldman Sachs Global Investment

Nickel, Lithium and Cobalt: The drivers of transport electrification

Last year, green demand accounted for >90% of demand growth for battery metals. Driven by strong China EV sales (+91% y/y) and battery capacity expansions (+ 50% y/y GSe), green lithium demand is estimated to have increased by 138kt LCE (+52% y/y) in 2022, 1.6 times the total green battery demand in 2020 (121kt LCE). The rapid rise of LFP batteries deployment in China, accounting now for 58% of China EV sales, resulted in slower green demand growth for nickel (25% y/y, +59kt) and cobalt (14% y/y, +7kt) last year. Into 2023, our auto Equity analysts estimate global EV sales to further increase by 40%, despite a moderation in China (+46% vs. 90% in 2022) and a persistence of

low growth in Europe due to the cost of living crisis. As such, we expect lithium battery demand to grow by 50%/y/y (+204kt LCE), cobalt by 32%/y/y (+18kt) and nickel by 35%/y/y (+104kt).

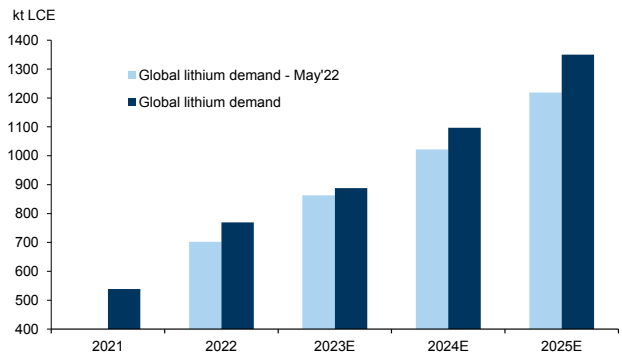
Exhibit 35: Increasing EV penetration is the main driver of green demand for battery metals
Battery metals green demand



*Lithium is in kt LCE

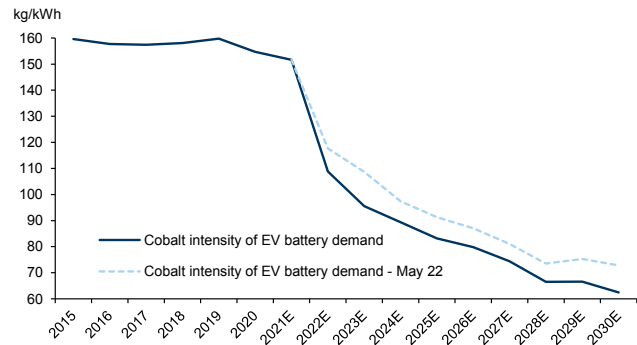
Source: Goldman Sachs Global Investment Research

Exhibit 36: Increasing policy commitment to EVs will sequentially increase lithium demand in the next years



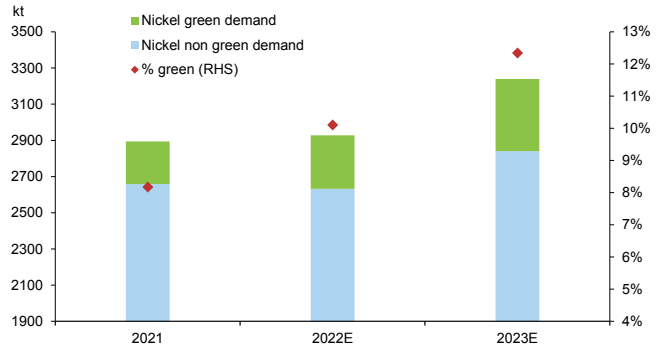
Source: Goldman Sachs Global Investment Research

Exhibit 37: Lower European EV sales and higher LFP share have contributed to declining EV demand cobalt intensity



Source: BNEF, Woodmac, Goldman Sachs Global Investment Research

Exhibit 38: Despite accounting for 12% of total demand, green sectors accounts for 33% of growth into 2023



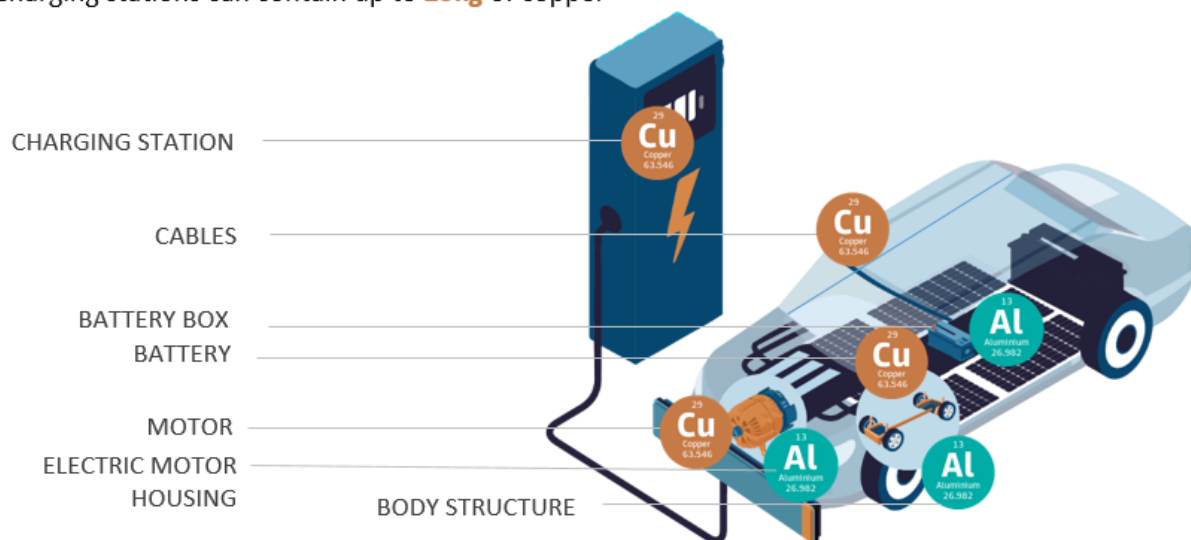
Source: Woodmac, Goldman Sachs Global Investment Research

Exhibit 39:

ELECTRIC VEHICLES

A pure electric vehicles contains up to **80kg** of copper (3.2x ICE vehicle content) and **227kg** of aluminium (1.5x ICE vehicle content).

Charging stations can contain up to **25kg** of copper



Source: World Bank, ICA, Copper Alliance, Goldman Sachs Global Investment Research

Exhibit 40: GS Lithium Green Demand Model

GS Lithium Green Demand Model										
'000 tonnes LCE	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
EV	215	331	482	682	863	1050	1193	1401	1696	2072
ESS	18	33	62	89	116	140	160	179	202	228
E-buses, two-wheeler EVs	34	42	66	86	102	120	146	187	235	263
Global green demand	267	406	609	857	1081	1310	1499	1767	2133	2564
YoY growth		52%	50%	41%	26%	21%	14%	18%	21%	20%
Europe	40	46	64	95	132	181	234	304	377	459
US	20	25	51	93	166	253	292	337	397	474
China	155	271	401	543	638	706	761	842	984	1186
RoW	19	21	28	40	44	50	67	97	139	181

Source: Goldman Sachs Global Investment Research

Exhibit 41: GS Cobalt Green Demand Model

GS Cobalt Green Demand Model										
'000 tonnes	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
EV	44	49	62	83	98	116	123	131	159	183
ESS	2	4	7	8	10	10	11	11	12	13
E-buses, two-wheeler EVs	3	2	3	5	5	6	7	8	8	10
Global green demand	48	55	72	96	113	132	141	150	179	205
YoY growth		14%	32%	33%	17%	17%	7%	6%	19%	15%
Europe	8	7	8	11	15	19	23	27	34	39
US	4	4	6	11	18	27	29	30	36	40
China	30	39	51	65	70	75	75	76	89	101
RoW	4	3	4	5	5	5	7	9	13	15

Source: Goldman Sachs Global Investment Research

Exhibit 42: GS Nickel Green Demand Model

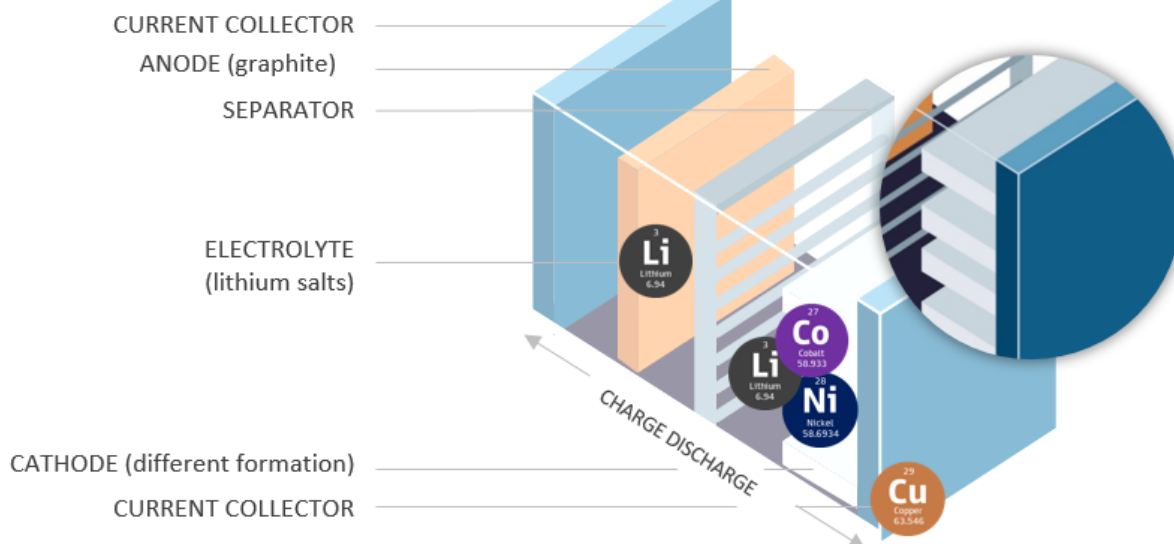
GS Nickel Green Demand Model										
'000 tonnes	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
EV	178	216	297	444	560	709	793	922	1122	1330
ESS	5	13	22	26	35	42	46	54	62	64
E-buses, two-wheeler EVs	18	29	40	47	58	62	74	82	86	99
Wind	35	39	42	48	56	62	68	74	78	83
Global green demand	237	296	400	565	710	875	981	1132	1348	1575
YoY growth	70%	25%	35%	41%	26%	23%	12%	15%	19%	17%
% of global demand	11%	13%	16%	21%	26%	31%	32%	36%	42%	46%
Europe	39	39	48	71	97	136	173	219	268	314
US	22	22	35	65	113	175	197	227	266	308
China	151	212	290	394	461	522	558	614	713	826
RoW	24	22	27	34	38	42	53	73	101	126

Source: Goldman Sachs Global Investment Research

Exhibit 43:

EV & ENERGY STORAGE BATTERIES

A 60kWh NMC 811 EV battery contains up to **50kg** of nickel, **9kg** of lithium and **6kg** of cobalt



Source: World Bank, ICA, Copper Alliance, IEA, Goldman Sachs Global Investment Research

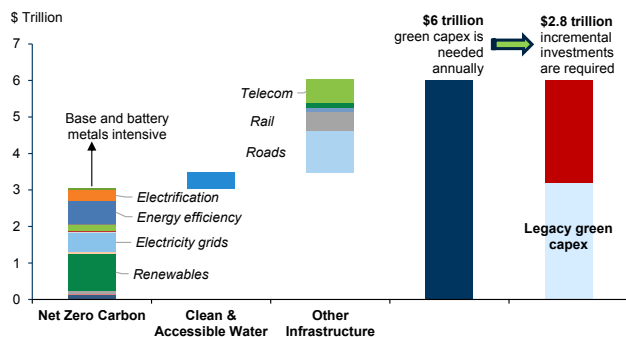
Accelerating Policy Support and Rise of the Green Capex

Alongside the gas-crisis-led increase in coal-fuelled energy consumption, 2022 saw a sharp increase in renewable installations and investments. Europe's solar and wind generated 22% of total electricity (from 19% in 2021), whilst China's share of renewable capacity reached 76% of total newly installed power generation capacity in 2022. New policies, such as the [Inflation Reduction Act](#) and [RePowerEU](#), are setting the stage for a renewables and EVs adoption acceleration. On the back of stronger policy targets and [improving economies of scale](#), we now expect on average 1.1Mty more copper demand

in the next 8 years (vs. our Apr-21 model) (see [Exhibit 26](#)). The sequential upgrade in green demand across metals is only starting, in our view, as countries try to converge to the path to net zero. Indeed, almost none of the countries are currently on track to reach the 1.5°C Paris Agreement target, and any additional step in that direction will increase global green metals demand. We estimate that under a net zero emission scenario, green copper demand would require on average an additional 2Mty between 2022/30 vs. our base case (see [Exhibit 45](#)).

Exhibit 44: \$2.8 trillion in annual investment is needed to meet zero carbon needs this decade

Annual investment required, 2020-2030

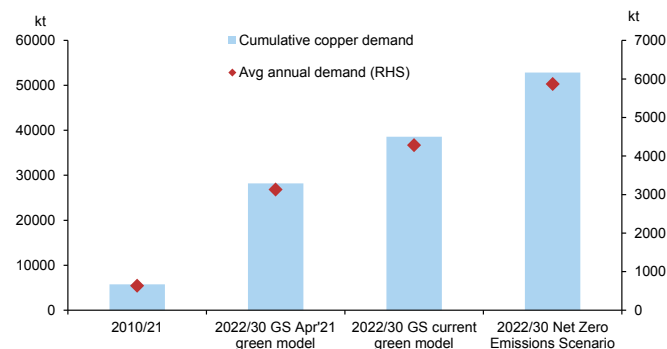


From the GS Sustain note: Green Capex: Making infrastructure happen

Source: IEA, OECD, McKinsey & Company, Goldman Sachs Global Investment Research

Exhibit 45: Upside risk to demand... Reaching Net Zero would require additional 2Mty of copper

Green demand from solar, wind & EVs



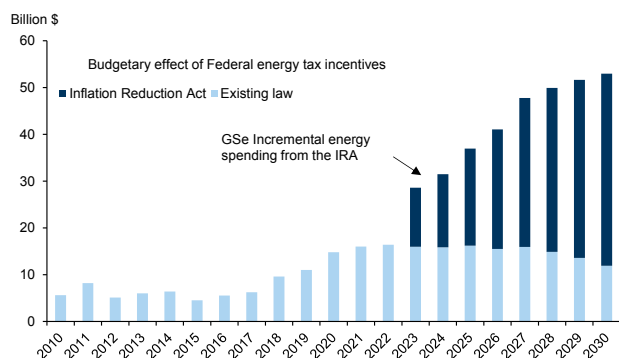
The IEA Net Zero Emissions by 2050 Scenario is a pathway for the global energy sector to achieve net zero CO2 emissions by 2050.

Source: IEA, Goldman Sachs Global Investment Research

United States: Inflation Reduction Act — a transformational change

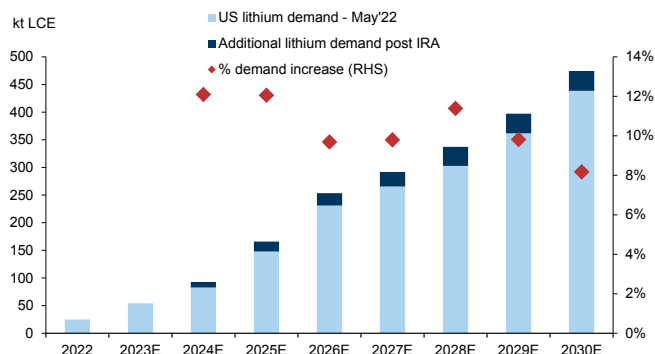
For the US, we view the Inflation Reduction Act (IRA) as a highly supportive measure for green capex and development of the green sectors. The incremental tax incentives – about \$260bn for energy transition (vs. \$80bn in Bipartisan Infrastructure law) – will be the driver of higher solar and wind deployments, battery storage increments and domestic manufacturing along with EV adoption acceleration. Our Equity analysts expect the extension in tax credit to double the total US federal tax incentives on energy and to drive US solar demand higher by 30% in 2025E (see [Exhibit 46](#)). We now expect 180kty (0.6% of global demand) and 500kty (0.65% of global demand) higher copper and aluminium demand on the back on the IRA between 2023-2030 (see [Exhibit 48](#)). Amongst battery metals, we expect a higher EV adoption to add 30kty LCE lithium demand annually (see [Exhibit 47](#)). The IRA will accelerate innovation and investment which will provide support for long-term green transition trends in the US, in our view.

Exhibit 46: IRA is estimated to increase energy spending by \$264 bn over the next 8 years



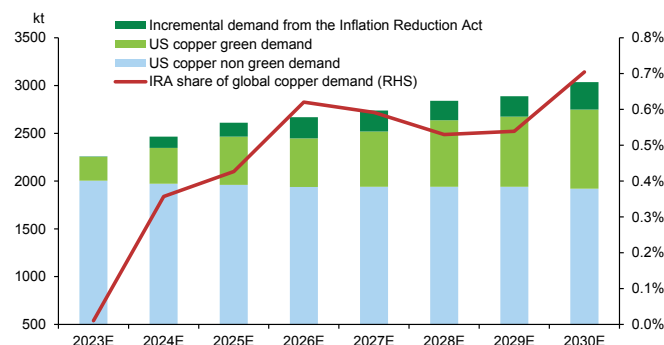
Source: Goldman Sachs Global Investment Research

Exhibit 47: The Inflation Reduction Act is supportive of US EV penetration increasing demand for battery metals
Based on GS model estimates



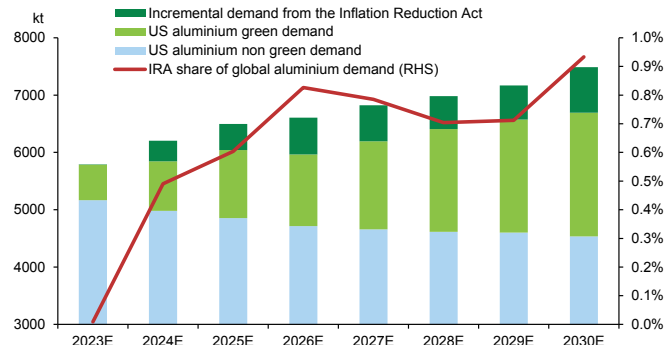
Source: Goldman Sachs Global Investment Research

Exhibit 48: Increasing incentives for green transition will lead to an additional 180kty copper



Source: Goldman Sachs Global Investment Research

Exhibit 49: The IRA will boost aluminium demand by more than 1% global demand by 2030



Source: Goldman Sachs Global Investment Research

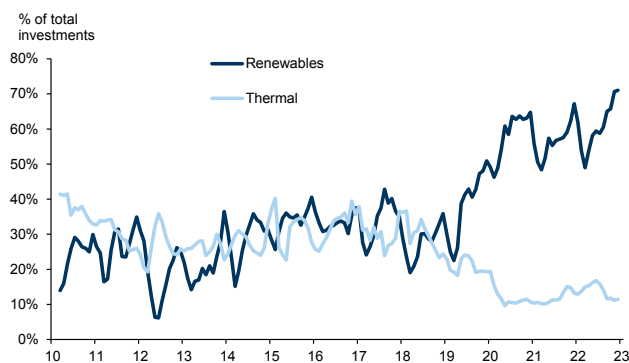
China: Aggressive renewable installation and ESS plans

Whilst last year China let fossil sources flourish, mainly coal, as it secured energy supply, it has continued to strengthen its commitment towards the green transition. Green capex in China remained strong with investments in renewables up by +42% y/y in 2022, accounting for 66% of total investments in power supply. Electric vehicles have remained supported by stimulus measures with EV penetration increasing sharply to 33% in 2022 (vs 17% in 2021). We expect Beijing to remain focussed on the green sector with the recent China Party Congress reiterating transition targets. Indeed, our Equity analysts have upgraded their forecasts for China's solar installation, driven by a sharp decline in poly prices. Furthermore, national targets for renewables currently stand at 1200GW by 2030, yet our Equity analysts expect this to be reached 5 years in advance. This is due to the continued decrease in renewables' LCOEs, grid parity and aggressive renewables installation plans at province and city levels, which are expected to drive 892GW of additional renewable installations between 2022-25, equal to 900Mty of copper and 1.1Mty of aluminium, and reach 1526GW of capacity in 2025. The accelerated renewables development is also expected to increase energy storage installations by 90% on average in the same period, leading to an additional 184 kt LCE

of lithium (11% of total demand) between 2023-25E. Over the long term, China's path stated policies will require 2.4Mty of copper (12% of annual Chinese demand), 4.4Mty of aluminium (10% of annual Chinese demand) and 900kty LCE of lithium between 2026-2030.

Exhibit 50: Despite weak growth and increased coal investments, green capex has remained strong

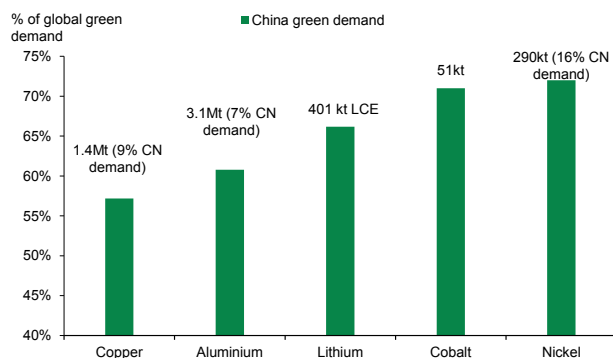
Renewables and thermal supply investments share over total power supply investments



Source: Wind, Goldman Sachs Global Investment Research

Exhibit 52: China represents 50-70% of green metals demand across the complex

2023 China share of global green demand & China green demand volumes, in kt*

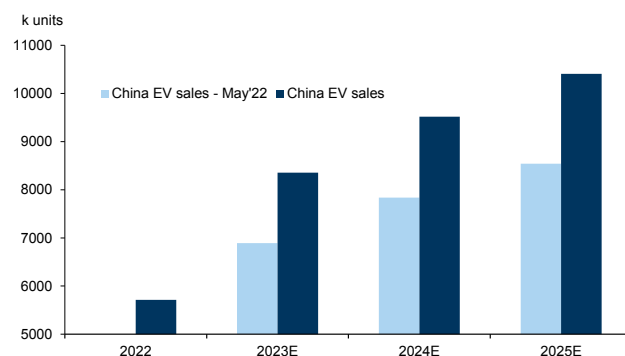


*Lithium demand volumes are in kt LCE

Source: Goldman Sachs Global Investment Research

Exhibit 51: China's EV sales GSe are boosted by >1mn in 2023 due to the outsized policy stimulus

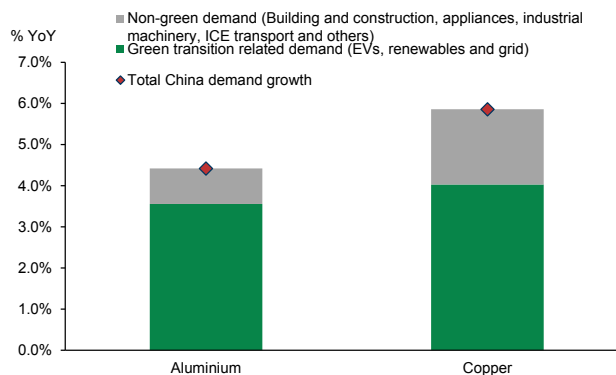
EV sales include PHEV and BEV



Source: Goldman Sachs Global Investment Research

Exhibit 53: Green and grid sectors are driving Chinese metals demand growth into 2023

Copper and Aluminium GS SD model



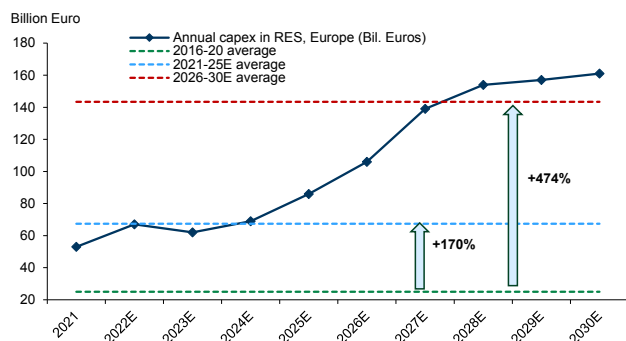
Source: Goldman Sachs Global Investment Research, Woodmac

Europe: RePowerEU, Renewables and Energy Security

In the aftermath of Russia's invasion of Ukraine, the European Union has released the RePowerEU plan which helps provides >€1 trn in investments in RES by 2030, and c.€0.7 trn in power networks (see Exhibit 54). And in response to the US Inflation Act, a "Net Zero Industry Act" (NZIA) is in formation which could make RePowerEU more concrete and actionable as our Equity analysts highlight, strengthening green demand support for metals. Indeed, solar and wind energy are key to the EU's push for energy independence from Russian fossil fuels and the acceleration of the green transition (see Exhibit 55). As a result of increased spending, we now expect EU renewables copper

demand at 454kty (7% of EU demand) over 2023-2030 and aluminium at 494kty (11% of EU demand). But if RePowerEU RES targets are implemented, the boost would be 1.5% and 2% in annual EU demand for copper and aluminium, respectively (see [Exhibit 57](#)). Whilst the energy crisis has boosted renewables, it has weighed on EU EV sales in the near term due to the cost-of-living crisis - our auto Equity analysts expect EU EV sales to amount to 3.1Mil (vs. 4.5Mil prev.) in 2023, resulting in a demand loss of 107kt of copper and 257kt of aluminium. However, the longer-term prospects for EVs in the EU remain strong as countries near their ICE phaseout deadlines and ambitions to transition remain high. Furthermore, second-round effects of renewables on energy storage and the power network will influence copper demand the most, given its higher intensity of use in digital grid and battery storage.

Exhibit 54: REPowerEU would imply >€1 trn in investments in RES by 2030, and c.€0.7 trn in power networks



Source: Goldman Sachs Global Investment Research

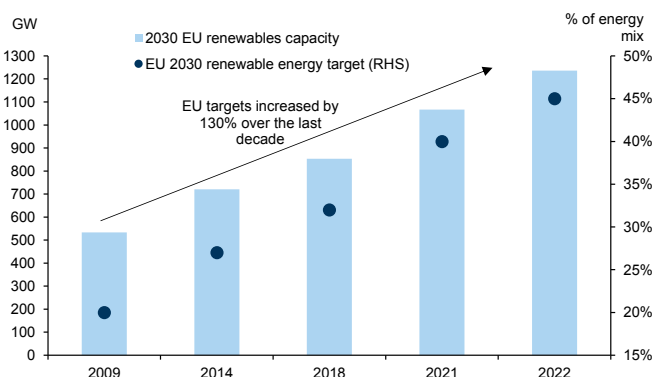
Exhibit 56: EU's renewable instalments outlined by RePowerEU this decade will require c.5Mt tonne of copper and 3.5Mt of aluminium



We assume the required 1236GW needed for 45% renewable share are achieved with solar & wind

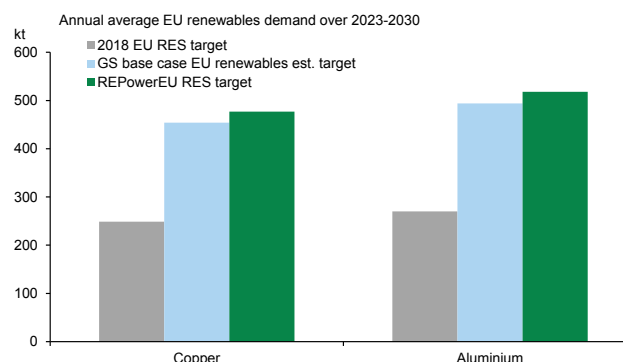
Source: Goldman Sachs Global Investment Research, IRENA, IEA, BNEF

Exhibit 55: Increasing EU 2030 renewables target has boosted green metals demand
2030 EU renewable energy target (%) and implied renewables capacity (RHS)



Source: European Commission, Goldman Sachs Global Investment Research

Exhibit 57: Full compliance with the REPower EU would bring demand to 500kty for copper and aluminium



Source: Goldman Sachs Global Investment Research

Green Supply Chains Dominated by China

Presently, the supply chains for the green transition are dominated by China. Around

60% of all green metals refining capacity, except copper which is at 40%, is located in China (see Exhibit 59), and the country also accounts for >70% of solar and battery supply chain, and >50% of wind turbines and nacelles. The desire for decarbonisation, recent geopolitical tensions and resulting push for energy independence have increased the focus on supply chain localisation. Green metals are central to the future of mobility and the green transition, with EV penetration rates now expected to reach 82 % and 70% in EU and US, respectively, by 2030. Amongst battery metals, we find the lithium supply chain as the low-hanging fruit in terms of relocation, compared to more concentrated and mature markets for nickel and cobalt. However, decoupling from China remains a challenge, given the economies of scale and relative gap in experience in the West. To compensate for these factors, higher capex flows and regulatory support will be required to develop ex-China supply chains. Indeed, the Inflation Reduction Act and Bipartisan Infrastructure funding in the US and the Green Deal in the EU incentivise supply chain localisation (see Exhibit 58). Finally, we see recycling and scrap playing major roles in the ex-China green metals supply chain. The funding from the DOE (US Department of Energy) for new recycling capacity development is the right step in that direction.

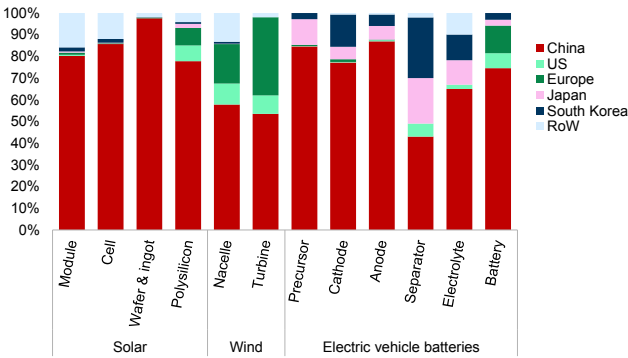
Exhibit 58: China controls the dominant share of refining capacity across green metals...

	Nickel	Cobalt	Copper	Aluminium
assets^	3%	3%	9%	21%
capacity	65%	68%	41%	58%
and	56%	70%	48%	59%

^ Based on 2023E

Source: Goldman Sachs Global Investment Research, Woodmac

Exhibit 59: ...and accounts for > 70% of solar and battery supply chain whilst >50% of wind.



Source: BNEF, IEA, Company data, Goldman Sachs Global Investment Research

Detailed Green Demand Models

Exhibit 60: GS Copper Green Demand Model

Global Green Copper Demand Model										
'000 tonnes	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
EU	357	447	484	607	759	1010	1227	1469	1653	1768
Solar	82	80	113	153	196	233	256	268	267	267
Wind	114	156	118	110	137	184	280	339	357	362
Stationary energy storage	2	3	4	5	6	7	8	9	10	10
Electric vehicles	153	199	235	320	397	555	644	806	963	1062
Charging infrastructure	6	9	13	18	24	31	39	47	57	67
US	162	208	257	500	661	743	815	923	976	1150
Solar	55	82	103	106	111	117	124	131	137	144
Wind	53	42	18	65	84	69	35	61	32	58
Stationary energy storage	5	7	11	14	17	20	23	26	31	35
Electric vehicles	46	75	120	308	439	524	615	681	747	877
Charging infrastructure	2	3	4	7	10	13	18	23	29	36
China	773	949	1486	1775	2028	2136	2257	2402	2555	2720
Solar	166	251	471	550	612	644	675	707	738	769
Wind	366	251	356	467	579	637	701	757	817	858
Stationary energy storage	2	6	14	23	32	39	45	49	55	62
Electric vehicles	228	426	625	710	774	779	793	841	890	968
Charging infrastructure	10	15	20	26	31	37	43	49	55	62
RoW	324	414	373	469	553	617	703	811	959	1146
Solar	222	287	249	339	363	389	415	456	504	553
Wind	86	106	95	91	119	117	129	131	142	165
Stationary energy storage	1	1	3	3	4	4	5	5	6	7
Electric vehicles	14	19	25	33	64	102	148	210	295	406
Charging infrastructure	1	1	2	3	4	5	7	9	11	15
Global green copper demand	1615	2019	2599	3350	4002	4506	5002	5604	6143	6784

By Region										
Green Demand (kt)	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
EU	357	447	484	607	759	1010	1227	1469	1653	1768
y/y growth		25%	8%	25%	25%	33%	21%	20%	13%	7%
US	162	208	257	500	661	743	815	923	976	1150
y/y growth		29%	23%	95%	32%	12%	10%	13%	6%	18%
China	773	949	1486	1775	2028	2136	2257	2402	2555	2720
y/y growth		23%	57%	19%	14%	5%	6%	6%	6%	6%
RoW	324	414	373	469	553	617	703	811	959	1146
y/y growth		28%	-10%	26%	18%	12%	14%	15%	18%	19%
Global	1615	2019	2599	3350	4002	4506	5002	5604	6143	6784
y/y growth		25%	29%	29%	19%	13%	11%	12%	10%	10%

By technology										
Green Demand (kt)	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Solar	526	701	936	1147	1282	1383	1470	1561	1646	1733
y/y growth		33%	34%	23%	12%	8%	6%	6%	5%	5%
Wind	619	554	587	733	919	1006	1145	1288	1349	1443
y/y growth		-10%	6%	25%	25%	10%	14%	12%	5%	7%
Stationary energy storage	10	17	31	46	59	70	81	89	101	115
y/y growth		82%	79%	47%	29%	19%	15%	10%	14%	13%
Electric vehicles	441	718	1005	1371	1674	1961	2201	2538	2895	3313
y/y growth		63%	40%	36%	22%	17%	12%	15%	14%	14%
Charging Infrastructure	19	29	40	53	68	86	106	128	153	180
y/y growth		49%	40%	33%	28%	26%	24%	21%	19%	18%
Total Green Demand	1615	2019	2599	3350	4002	4506	5002	5604	6143	6784
y/y growth		25%	29%	29%	19%	13%	11%	12%	10%	10%

Source: Goldman Sachs Global Investment Research

Exhibit 61: GS Aluminium Green Demand Model

Global Green Aluminium Demand Model										
'000 tonnes	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
EU	582	692	827	1077	1327	1746	1992	2359	2687	2878
Solar	175	170	239	323	415	493	542	568	568	569
Wind	14	17	15	17	20	26	35	39	41	42
Electric vehicles	392	504	573	738	891	1226	1415	1752	2078	2267
US	265	380	533	987	1308	1501	1712	1872	2028	2325
Solar	140	200	253	263	276	296	316	336	356	377
Wind	12	9	4	7	9	8	6	8	6	7
Electric vehicles	113	171	276	717	1023	1197	1390	1528	1666	1940
China	1127	1968	3115	3586	3958	4056	4180	4399	4620	4922
Solar	398	600	1125	1313	1463	1538	1613	1688	1763	1838
Wind	42	45	56	70	83	91	100	108	117	123
Electric vehicles	688	1323	1934	2204	2412	2428	2467	2603	2741	2962
RoW	565	733	649	889	1025	1162	1319	1546	1842	2196
Solar	512	669	572	796	859	920	980	1078	1192	1307
Wind	17	17	19	17	19	19	19	19	20	23
Electric vehicles	35	47	59	76	147	223	320	449	630	867
Global	2539	3773	5124	6539	7617	8465	9202	10176	11177	12321

By Region Green Demand (kt)	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
EU	582	692	827	1077	1327	1746	1992	2359	2687	2878
y/y growth		19%	20%	30%	23%	32%	14%	18%	14%	7%
US	265	380	533	987	1308	1501	1712	1872	2028	2325
y/y growth		44%	40%	85%	33%	15%	14%	9%	8%	15%
China	1127	1968	3115	3586	3958	4056	4180	4399	4620	4922
y/y growth		75%	58%	15%	10%	2%	3%	5%	5%	7%
RoW	565	733	649	889	1025	1162	1319	1546	1842	2196
y/y growth		30%	-11%	37%	15%	13%	14%	17%	19%	19%
Global	2539	3773	5124	6539	7617	8465	9202	10176	11177	12321
y/y growth		49%	36%	28%	16%	11%	9%	11%	10%	10%

By technology Green Demand (kt)	2022E	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Solar	1225	1640	2189	2693	3012	3247	3451	3670	3879	4090
y/y growth		34%	33%	23%	12%	8%	6%	6%	6%	5%
Wind	86	88	94	110	131	144	159	174	183	195
y/y growth		3%	7%	17%	19%	10%	11%	9%	6%	6%
Electric vehicles	1228	2045	2842	3736	4474	5074	5591	6333	7115	8036
y/y growth		67%	39%	31%	20%	13%	10%	13%	12%	13%
Total Green Demand	2539	3773	5124	6539	7617	8465	9202	10176	11177	12321
y/y growth		49%	36%	28%	16%	11%	9%	11%	10%	10%

Source: Goldman Sachs Global Investment Research

Exhibit 62: GS Cobalt Green Demand Model

GS Cobalt Green Demand Model										
000 tonnes	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Europe	8	7	8	11	15	19	23	27	34	39
EV	8	6	7	10	13	18	21	25	32	36
ESS	0	1	1	1	1	2	2	2	2	3
YoY growth	96%	-14%	21%	40%	28%	32%	21%	18%	24%	15%
US	4	4	6	11	18	27	29	30	36	40
EV	4	3	6	10	17	25	27	28	33	38
ESS	0	0	1	1	2	2	2	2	2	3
YoY growth	54%	-5%	78%	71%	66%	47%	8%	5%	18%	13%
China	30	39	51	65	70	75	75	76	89	101
EV	29	36	46	59	64	69	69	70	83	94
ESS	1	3	5	6	6	6	6	6	6	7
YoY growth	162%	31%	30%	26%	9%	6%	1%	0%	17%	14%
RoW	4	3	4	5	5	5	7	9	13	15
EV	4	3	3	4	4	5	6	8	12	14
ESS	0	0	0	0	0	0	1	1	1	1
YoY growth	50%	-17%	18%	32%	1%	9%	26%	31%	44%	23%
Global cobalt green demand	48	55	72	96	113	132	141	150	179	205
EV	44	49	62	83	98	116	123	131	159	183
ESS	2	4	7	8	10	10	11	11	12	13
E-buses, two-wheeler EVs	3	2	3	5	5	6	7	8	8	10
YoY growth	116%	14%	32%	33%	17%	17%	7%	6%	19%	15%

Source: Goldman Sachs Global Investment Research

Exhibit 63: GS Lithium Green Demand Model

GS Lithium Green Demand Model										
'000 tonnes LCE	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Europe	40	46	64	95	132	181	234	304	377	459
EV	37	42	56	84	116	160	206	270	337	413
ESS	3	4	7	11	16	21	28	34	40	46
YoY growth	102%	15%	38%	50%	38%	37%	29%	30%	24%	22%
US	20	25	51	93	166	253	292	337	397	474
EV	18	23	45	82	146	224	257	299	355	427
ESS	2	2	6	11	20	30	35	38	42	47
YoY growth	59%	28%	102%	84%	78%	53%	15%	16%	18%	19%
China	155	271	401	543	638	706	761	842	984	1186
EV	143	247	356	480	562	623	671	746	880	1069
ESS	12	25	46	63	76	83	90	95	105	118
YoY growth	171%	75%	48%	35%	17%	11%	8%	11%	17%	21%
RoW	19	21	28	40	44	50	67	97	139	181
EV	17	19	25	36	39	44	59	86	124	163
ESS	1	2	3	5	5	6	8	11	15	18
YoY growth	55%	12%	35%	42%	9%	13%	35%	44%	44%	30%
Global lithium green demand	267	406	609	857	1081	1310	1499	1767	2133	2564
EV	215	331	482	682	863	1050	1193	1401	1696	2072
ESS	18	33	62	89	116	140	160	179	202	228
E-buses, two-wheeler EVs	34	42	66	86	102	120	146	187	235	263
YoY growth	120%	52%	50%	41%	26%	21%	14%	18%	21%	20%

Source: Goldman Sachs Global Investment Research

Exhibit 64: GS Nickel Green Demand Model

GS Nickel Green Demand Model

000 tonnes	2021	2022E	2023E	2024E	2025E	2026E	2027E	2028E	2029E	2030E
Europe	39	39	48	71	97	136	173	219	268	314
EV	33	31	39	61	84	119	151	193	240	284
ESS	1	2	3	3	5	6	8	10	12	13
Wind	5	7	6	7	9	11	14	16	16	17
YoY growth	65%	1%	22%	48%	36%	40%	27%	27%	22%	17%
US	22	22	35	65	113	175	197	227	266	308
EV	16	17	31	58	104	163	185	212	250	292
ESS	0	1	2	3	6	9	10	12	13	13
Wind	6	4	2	3	4	3	3	3	3	3
YoY growth	29%	-1%	59%	85%	76%	55%	13%	15%	17%	16%
China	151	212	290	394	461	522	558	614	713	826
EV	131	183	249	345	403	457	489	538	631	741
ESS	3	9	16	19	23	25	26	29	32	33
Wind	17	20	25	30	36	39	43	47	50	53
YoY growth	84%	40%	36%	36%	17%	13%	7%	10%	16%	16%
RoW	24	22	27	34	38	42	53	73	101	126
EV	16	14	17	25	28	32	43	61	88	112
ESS	0	1	1	1	2	2	2	3	5	5
Wind	8	7	9	8	8	8	8	8	8	10
YoY growth	54%	-8%	23%	27%	10%	13%	26%	36%	39%	25%
Global green demand nickel	237	296	400	565	710	875	981	1132	1348	1575
EV	178	216	297	444	560	709	793	922	1122	1330
ESS	5	13	22	26	35	42	46	54	62	64
E-buses, two-wheeler EVs	18	29	40	47	58	62	74	82	86	99
Wind	35	39	42	48	56	62	68	74	78	83
YoY growth	70%	25%	35%	41%	26%	23%	12%	15%	19%	17%

Source: Goldman Sachs Global Investment Research

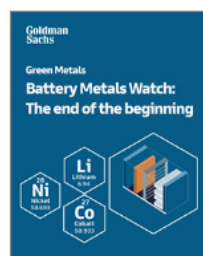
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Green Metals: Battery Metals Watch: Lithium's delayed decline

9 Nov 2022

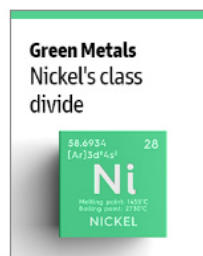
In Q2 this year, we called for peaking battery metals prices as mounting supply responses were set to catch up to the accelerating demand. Since then, the cobalt price has fallen 35% and nickel is down 40%, but onshore lithium prices have risen (24%) to record new levels. For lithium, whilst the supply response we had estimated has continued to gather pace, as evidenced by surging Chinese lithium carbonate output, LFP battery expansion related restocking demand and higher-than-expected EV sales in China due to the exceptional stimulus have kept the market tighter than expected.



Green Metals: Battery Metals Watch: The end of the beginning

29 May 2022

Battery metals – cobalt, lithium and nickel – will power the green industrial revolution, facing a wave of demand comparable to that of copper and iron ore during China's rapid growth in the 2000's. With climate change top of mind, investors are fully aware that battery metals will play a crucial role in the 21st century global economy, just as bulk and base metals did before them. Yet despite this exponential demand profile, we see the battery metals bull market as over for now.



Green Metals: Nickel's class divide

28 Apr 2022

At the beginning of the year, nickel's place within green metals was as a key competitor in the race for mineral dominance of energy storage. Now, it sits at the intersection of Europe's push for decarbonisation and energy independence. At the heart of Europe's strategy lies its desire to rapidly electrify its transportation sector – a source of 20% of its emissions and c.2mb/d of Russian oil imports.



Green Metals: Solving Aluminium's Climate Paradox

20 Jun 2021

At the heart of the coming surge in green aluminium demand lies a paradox: aluminium is a key input required to produce decarbonising technologies like EV's and solar power, yet its own production is very carbon intensive, generating 2% of all global emissions. This paradox begs the question: how can we secure enough aluminium to effectively decarbonise, while keeping the climate impact of the path to net zero to a minimum? In our view, the resolution of this paradox will drive a structural bull market in aluminium over the next half decade, driven by the necessity to grow supply to meet green demand while cutting emissions to prevent a climate catastrophe.



Green Metals: Copper is the new oil

13 Apr 2021

The critical role copper will play in achieving the Paris climate goals cannot be overstated. As the most cost-effective conductive material, copper sits at the heart of capturing, storing and transporting these new sources of energy. Leveraging equity analysis across EVs, wind, solar, and battery technology, we estimate that by 2030, copper demand will grow nearly 600%. Yet the copper market as it currently stands is not prepared for this demand, and producers are leery of committing to new supply. We significantly increase our price forecasts and explore the implications across the metals and mining complex.

Disclosure Appendix

Reg AC

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